

# New T&M Tools for Academia Boost Learning, R&D and Industry Readiness



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**T**est and measurement (T&M) companies have been serving the education vertical for a very long time now. The education sector consumes a variety of products ranging from basic instruments, like power and signal sources, to oscilloscopes, analysers and digital multimeters. T&M instruments are vital to any electronics lab.

Apart from being high on the ease-of-use and learnability factor, today's latest T&M equipment provide a platform for students to bring their imaginative ideas to life and develop their skills right from basic engineering to advanced level. Available at pocket-friendly prices to expensive but highly featureful, these instruments also aid academicians in their research and development (R&D) activities as well as performing and writing up new experiments. Let us take a look at the current trends and latest arrivals in the T&M domain targeting colleges and universities.

## Emphasis on adapting industry-grade products for academia

Industry experts feel that the trend of products singularly manufactured for academia has decreased over the years. Although some

of the T&M companies produce instruments particularly targeting the education sector, others are focusing on pitching industry-grade products for academia, be it oscilloscopes, spectrum analysers, signal generators or even digital multimeters today.

Employing the same full-featured set of T&M tools operated by professionals helps students to comfortably transition into the industry without having to start from scratch. Talking from National Instruments' philosophical standpoint, Satish Mohanram, technical marketing manager, National Instruments, says, "We believe in teaching engineering through experiential learning. Therefore we feel there should be no separation between T&M tools used by academia or industry."

He adds, "When we come up with a product, apart from it being used by the industry, we consider in getting students trained on that tool as they learn their engineering so that eventually when they graduate, they do not have to start from scratch in learning new tools to become productive. From that standpoint, we have been launching quite a few products that significantly affect the industry and could

be used in the academic scenario to teach some really important concepts very effectively."

For academia, Keysight Technologies provide the same equipment that they provide industries for T&M measurement purposes. There is no specific focus on products for academia. Pathy V.V., business development manager-education, Keysight Technologies India Pvt Ltd, says, "Academia is just one of the markets. We have launched products for all domains." He adds, "We identify what makes sense for education and then position those products for educational institutes."

## View point: Why academia should switch to digital storage oscilloscopes and use industry-grade T&M instruments

Today, I find a lot of colleges still using analogue oscilloscopes. We are advising them to switch to digital storage oscilloscopes with at least 50MHz bandwidth. If you look at oscilloscope measurements today, no industry uses analogue mode. As a matter of fact, there are hardly any analogue radios visible today.

Also, the reason for not focusing on making products particularly for academia is because we want them to use the very same equipment that they will see when employed in an industry. When they enter the industry, they should be able to immediately use that equipment without starting from scratch. When we set up labs in academic institutes, we offer the very same model but they get a benefit in price. By using the same equipment, budding engineers become eligible for technical jobs.

Depending on academic needs, basic oscilloscopes are available for undergraduate teaching and advanced research, and oscilloscopes with high bandwidths are preferred.

—Pathy V.V., business development manager-education,  
Keysight Technologies India Pvt Ltd